Experiment:4

Generation and Analysis of Frequency Modulated (FM) and of Phase Modulated (PM) Signals using MATLAB

Aim: To generate Frequency Modulated (FM) and Phase Modulated (PM) signals using MATLAB. To analyze their time-domain waveforms, frequency spectra, and modulation indices.

Software Required: MATLAB

Program:

```
clc;
clear;
close all;
% Generation and Analysis of FM and PM Signals Using MATLAB
%% Step 1: Define Parameters
Fs = 100000;            % Sampling frequency in Hz
T = 0.02;               % Simulation duration in seconds
t = 0:1/Fs:T-1/Fs;      % Time vector
N = length(t);
Ac = 5;                 % Carrier amplitude in volts
Am = 1;                 % Message amplitude in volts
fc = 10000;             % Carrier frequency in Hz
fm = 500;               % Message frequency in Hz
delta_f = 2000;         % Peak frequency deviation for FM (Hz)
Kp = 2;                 % PM sensitivity (radians per volt)
%% Step 2: Generate Message and Carrier Signals
```

```
message = Am*cos(2*pi*fm*t);
carrier = Ac*cos(2*pi*fc*t);
%% Step 3: Generate Frequency Modulated (FM) Signal
% FM modulation index
beta_FM = delta_f/fm;

% FM signal
FM_signal = Ac*cos(2*pi*fc*t + ...
            beta_FM*sin(2*pi*fm*t));
%% Step 4: Generate Phase Modulated (PM) Signal
% PM modulation index
beta_PM = Kp*Am;
% PM signal
PM_signal = Ac*cos(2*pi*fc*t + ...
            Kp*message);
%% Step 5: Calculate Instantaneous Frequencies
FM_frequency = fc + delta_f*cos(2*pi*fm*t);
PM_frequency = fc - Kp*Am*fm*sin(2*pi*fm*t);
%% Step 6: Display Results
fprintf('\nFM AND PM SIGNAL ANALYSIS\n');
fprintf('------------------------\n');
fprintf('Carrier frequency       = %.2f Hz\n', fc);
fprintf('Message frequency       = %.2f Hz\n', fm);
```

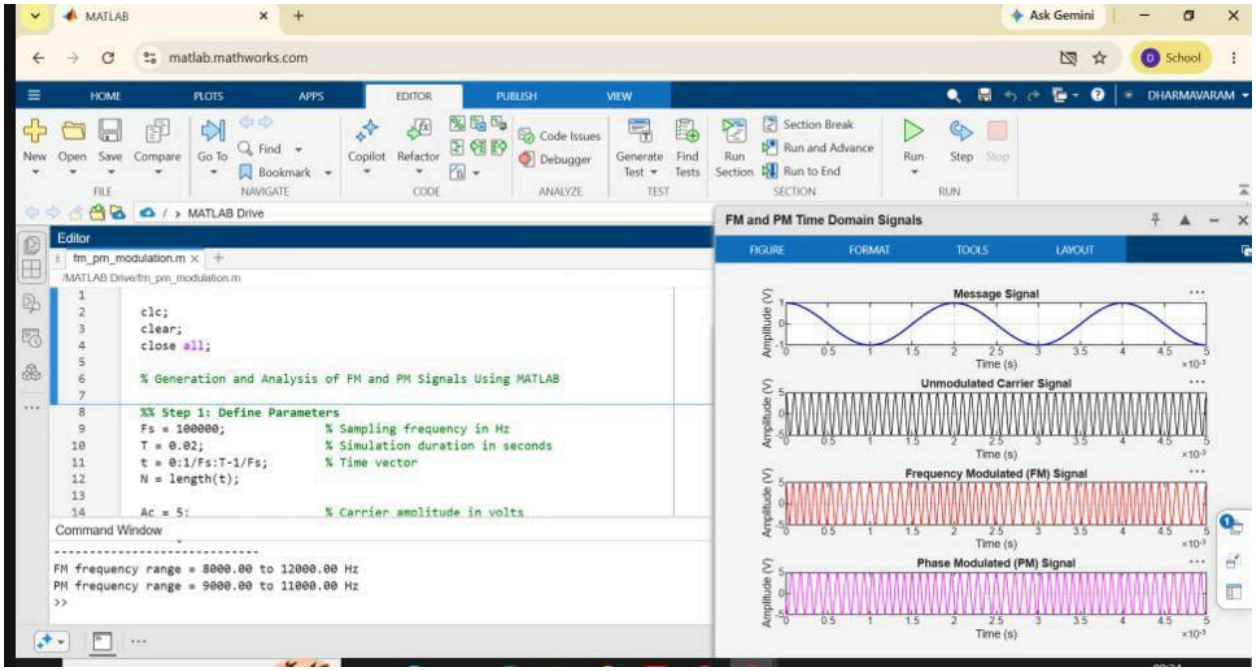
```
fprintf('Carrier amplitude       = %.2f V\n', Ac);
fprintf('Message amplitude       = %.2f V\n', Am);
fprintf('FM frequency deviation  = %.2f Hz\n', delta_f);
fprintf('FM modulation index     = %.2f\n', beta_FM);
fprintf('PM sensitivity          = %.2f rad/V\n', Kp);
fprintf('PM modulation index     = %.2f rad\n', beta_PM);
fprintf('FM minimum frequency    = %.2f Hz\n', fc-delta_f);
fprintf('FM maximum frequency    = %.2f Hz\n', fc+delta_f);
fprintf('PM frequency deviation  = %.2f Hz\n', Kp*Am*fm);
fprintf('Simulation completed successfully.\n');
%% Step 7: Plot Time-Domain Signals
figure('Name','FM and PM Time Domain Signals', ...
     'NumberTitle','off');
nPlot = round(0.005*Fs);
tp = t(1:nPlot);
subplot(4,1,1);
plot(tp, message(1:nPlot), 'b', 'LineWidth', 1.2);
grid on;
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude (V)');
subplot(4,1,2);
plot(tp, carrier(1:nPlot), 'k');
```

```
grid on;
title('Unmodulated Carrier Signal');
xlabel('Time (s)');
ylabel('Amplitude (V)');
subplot(4,1,3);
plot(tp, FM_signal(1:nPlot), 'r');
grid on;
title('Frequency Modulated (FM) Signal');
xlabel('Time (s)');
ylabel('Amplitude (V)');
subplot(4,1,4);
plot(tp, PM_signal(1:nPlot), 'm');
grid on;
title('Phase Modulated (PM) Signal');
xlabel('Time (s)');
ylabel('Amplitude (V)');
%% Step 8: Plot Instantaneous Frequency
figure('Name','Instantaneous Frequency Analysis', ...
    'NumberTitle','off');
subplot(2,1,1);
plot(tp, FM_frequency(1:nPlot), 'r', 'LineWidth', 1.2);
grid on;
title('Instantaneous Frequency of FM Signal');
```

```
xlabel('Time (s)');
ylabel('Frequency (Hz)');
subplot(2,1,2);
plot(tp, PM_frequency(1:nPlot), 'm', 'LineWidth', 1.2);
grid on;
title('Instantaneous Frequency of PM Signal');
xlabel('Time (s)');
ylabel('Frequency (Hz)');
%% Step 9: Calculate Frequency Spectra
f = (0:floor(N/2))*Fs/N;
FM_FFT = fft(FM_signal);
PM_FFT = fft(PM_signal);
FM_spectrum = abs(FM_FFT(1:floor(N/2)+1))/N;
PM_spectrum = abs(PM_FFT(1:floor(N/2)+1))/N;
% Convert to one-sided amplitude spectra
if mod(N,2) == 0
   FM_spectrum(2:end-1) = 2*FM_spectrum(2:end-1);
   PM_spectrum(2:end-1) = 2*PM_spectrum(2:end-1);
Else
  FM_spectrum(2:end) = 2*FM_spectrum(2:end);
   PM_spectrum(2:end) = 2*PM_spectrum(2:end);
end
%% Step 10: Plot Frequency Spectra
```

```
figure('Name','FM and PM Frequency Spectra', ...
    'NumberTitle','off');
subplot(2,1,1);
plot(f/1000, FM_spectrum, 'r', 'LineWidth', 1.2);
grid on;
title('Frequency Spectrum of FM Signal');
xlabel('Frequency (kHz)');
ylabel('Amplitude');
xlim([0 20]);
subplot(2,1,2);
plot(f/1000, PM_spectrum, 'm', 'LineWidth', 1.2);
grid on;
title('Frequency Spectrum of PM Signal');
xlabel('Frequency (kHz)');
ylabel('Amplitude');
xlim([0 20]);
%% Step 11: Display Peak Frequency Deviations
fprintf('\nINSTANTANEOUS FREQUENCY CHECK\n');
fprintf('----------------------------\n');
fprintf('FM frequency range = %.2f to %.2f Hz\n', ...
    min(FM_frequency), max(FM_frequency));
fprintf('PM frequency range = %.2f to %.2f Hz\n', ...
    min(PM_frequency), max(PM_frequency));
```

Output:



Result: FM and PM signals were successfully generated and analyzed using MATLAB. Their waveforms, instantaneous frequency variations, and frequency spectra were observed and verified.